

Transformer Based Music Transcription & MusicXML Generation

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PROBLEM & GOAL

Music transcription is hard because real audio mixes many instruments, timing variations, noise, and performance expression.



Mixed sources

vocals, bass, drums, piano, guitar overlap



Performance timing

human playing does not align perfectly to notation



Amateur Musicians

hard to find sheet music for the pieces they wish to perform



Usability

musicians need editable MusicXML, not only MIDI

Generate an editable first draft of sheet music from audio clips

Background

● MIDI

- A symbolic music format that stores musical events
- Includes information such as pitch, note start time, duration, and velocity
- Similar to a “digital performance record,” not actual audio

◆ MusicXML

- A notation-based file format for sheet music
- Stores notes, rests, measures, clefs, time signatures, key signatures, and instrument parts
- Can be opened and edited in notation software such as MuseScore

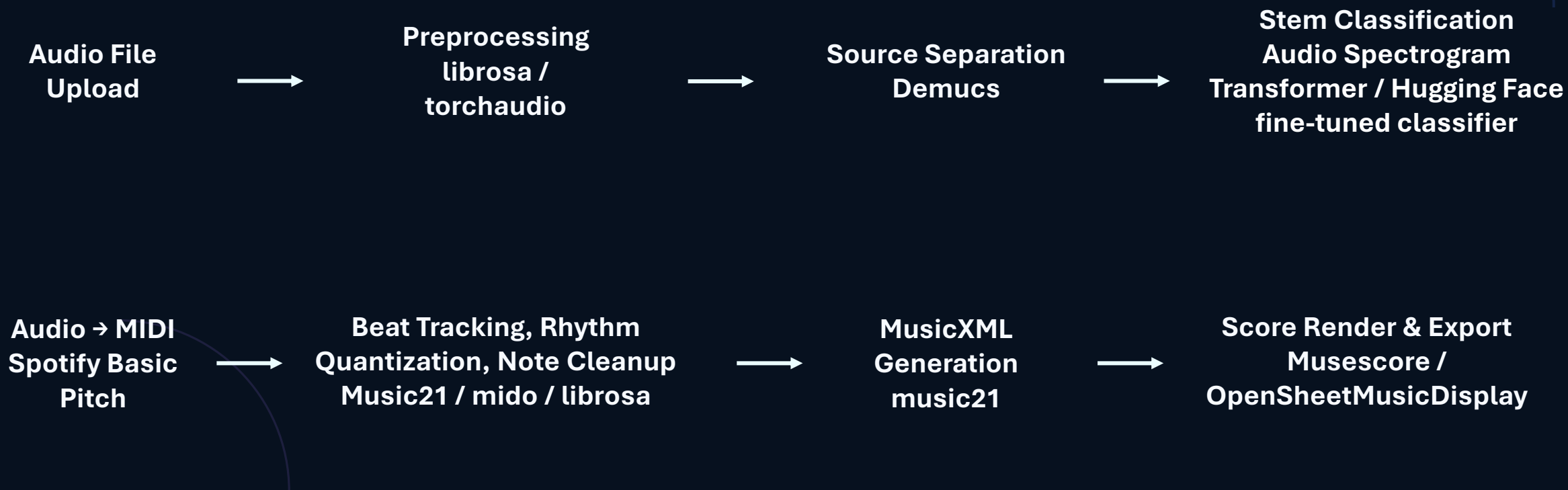
▀ Source Separation

- The process of splitting mixed audio into stems
- Example stems: vocals, drums, bass, and accompaniment

♪ Music Transcription

- The process of converting performed audio into symbolic notes
- In this project, transcription connects audio input to MIDI-like note data and then to MusicXML sheet music

PIPELINE



Core AI and signal-processing methods

Separate

**Demucs / Hybrid Transformer
Demucs**

Split mixed audio into various instruments like vocals, drums, bass, and accompaniment stems

Classify

**CNN / Audio Spectrogram
Transformer / YAMNet**

Label each stem as melody, bass, piano, guitar, drums or unknown

Transcribe

**Spotify Basic Pitch / ByteDance
piano_transcription**

Convert selected stems into MIDI-like pitch, onset, offset, duration, and velocity.

Core AI and signal-processing methods

Cleanup

Librosa, Madmom, music21, pretty_midi

Convert raw note events into readable notation with beat tracking, tempo estimation, onset alignment

MusicXML Generation

music21, partitura

Convert cleaned note events into score format

Render

OpenSheetMusicDisplay / MuseScore CLI

Users can view, edit, print, and reuse the generated score

Technology Stack

Backend

Python / FastAPI

PyTorch / TensorFlow

librosa / torchaudio / ffmpeg

Demucs + Basic Pitch

music21 / partitura / mido

MuseScore CLI

Frontend

React / TypeScript

OpenSheetMusicDisplay

Waveform + stem player

Correction interface

Expected Outcome

**A practical web-based music transcription assistant
that produces editable first-draft sheet music**

- Converts short audio recordings into editable sheet music
 - Uses Demucs-based source separation to isolate musical stems
 - Applies stem classification to identify instrument or musical role
 - Transcribes audio into MIDI-like note events using Basic Pitch
 - Supports optional piano-specific transcription for piano stems
 - Cleans raw transcription through rhythm quantization and symbolic processing
 - Generates MusicXML for editable notation
 - Renders scores with MuseScore and OpenSheetMusicDisplay
 - Creates a practical human-in-the-loop workflow for music transcription
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